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Project Title:	Catalyzed Combustion for Extra-Terrestrial Applications		
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Program/Discipline:	COMBUSTION SCIENCE		
Program/Discipline--Element/Subdiscipline:	COMBUSTION SCIENCE--Surface combustion, shs, fire safety and smoldering		
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Task Description:	In this task we research a catalytically stabilized, premixed flame in a jet configuration where the jet diameter is smaller than a typical macro-scale flame thickness. Both a nozzle stabilized flame with some of the combustion occurring inside of the catalytic delivery tube, as well as a stagnation plate and enclosed flame configuration, will be examined. The research will explore the potential for a microjet flame as a heater, a power source, or a torch capable of operation in the Martian atmosphere. The basic concept is that of a small, pre-mixed fuel/oxidizer flow that begins combusting inside a capillary-size (1 mm diameter or smaller) delivery tube, the outlet section of which is a pure catalyst, and then completes combustion either as a free jet flame or impinging on a flat plate as a stagnation flame. The plate can also be catalytic to increase combustion efficiency and reduce the chance of reaction quenching. Of particular fundamental interest will be the flame structure and stability when the outlet tube diameter is on the order of or less than a typical premixed flame thickness. The proposed program will consist of normal gravity experiments primarily in air and low-pressure carbon-dioxide atmospheres (i.e. Martian). Some exploratory work will investigate the feasibility of burning in a vacuum environment. The first stage of experiments will investigate the lower size limits of a premixed, catalyzed, partially-precombusted free jet flame, and gather fundamental combustion data on a scale where none currently exist. Fuels to be used include hydrogen, as well as fuels that may be made in-situ on Mars such as carbon monoxide, methane, and possibly methanol (all mixed with hydrogen to obtain the catalytic effect); the oxidizer will be oxygen partially diluted with an inert such as nitrogen or carbon dioxide to prevent catalyst burnout. The catalysts will be platinum and palladium. We will also measure the downstream combustion efficiency and its dependence on length of catalytic tube section (or the flow residence time in the catalytic section). In the second stage a stagnation plate will be added to the flow to obtain data on the effect of heat loss from the jet and to evaluate the potential of the jet as a microtorch. A catalytic plate will also be studied to determine the flame structure in a stagnation flow field. The third stage of experiments will occur later in the program and involve integration of the impinging jet flat flame burner into a more realistic enclosed combustor, which will allow high-pressure operation. In addition to fundamental information on the flame structure, the experiments will provide operating characteristics and limits of a pre-mixed microjet flame. The heat transfer to the stagnation plate is of interest for heater applications. The diagnostics for the experiments will include: 1. Standard and spectral imaging of the flame; 2. Temperature measurements of the solid plate and gas-phase flame using thermocouples, thin filament pyrometry and infrared imaging; 3. Particle Image Velocimetry measurements of the gas-phase flow profiles, and 4) Planar Laser Induced Fluorescence and Rainbow Schlieren measurements on the reacting flow field. Measurements will be made over a range of jet exit diameters, gas velocities, operating pressures, catalyst types, catalyst section lengths, external environments, plate-to-jet separation distances, equivalence ratios and types and percentages of diluent gases. The program will also involve the development of a detailed numerical model, using a commercial code, to predict the behavior of the combustion. The model will be two-dimensional, axi-symmetric, and solve the full Navier-Stokes equations for momentum and the energy and species conservation equations, including surface radiation. We will use detailed chemistry for both the gas and surface reactions. Currently, detailed mechanisms (including surface chemistry) for most of the candidate fuels and catalysts are available in the technical literature. The objective is to determine the fundamentals of catalyzed premixed jet flames at sub-millimeter scales. Quantities such as flame shape, flammability limits, temperature, acoustic emission, catalytic effects, etc. will be measured as a function of equivalence ratio, velocity, pressure, and fuel type. NASA is in need of simple, small, power systems, and combustion offers a largely untapped technology that may have many advantages over the batteries technology currently used. The high power density offered by combustion systems, along with easy recharge, make the consideration of micro premixed flames for heating or conversion to power attractive.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Microcombustors have the potential to supply personal power for a variety of devices with advantage over current power sources like batteries. There is a lack of fundamental data on the characteristics of small-scale premixed flames, and this project will supply data useful for designing devices and verifying models. The resulting flame would have a wide range of potential applications as a heater, torch, or the basis for a microcombustor electrical generator.
	Progress was achieved on both the experiment and the numerical model during this reporting period. We constructed an apparatus that allows the precise metering and mixing of oxygen and a fuel gas (methane or carbon monoxide) and the ignition of this mixture at the end of a small tube. A nitrogen co-flow shroud was built to shield the flame from the effects of ambient oxygen in the room. A camera with microscope lens recorded the flame size and shape, a microphone recorded the flame sound, and a video camera recorded the flame as it moved inside the tube. Three flame regimes were studied: a flame stabilized outside the tube, the flame stabilized just inside the end of the tube, and a traveling flame inside the tube. The tube diameter, wall thickness, and material were varied to determine their effects. For the flame stabilized on the outside of the tube, the blow-off limits were determined and flame shape and size recorded and measured over a range of tube sizes, equivalence ratios and mixture velocities. At low speed the flame forms a rounded cap on the tube outlet, while at higher speed the flame stretches out and more closely resembles a standard Bunsen flame. It appears that in some cases reaction continues to take place beyond the bright flame cone. The richer flames (e.g., phi = 1.3) move closer to the tube and heat it sufficiently that it glows slightly due to thermal emission. This may be due to air diffusing to the flame and providing extra oxygen. In the nitrogen shroud, the flame extinguished above phi = 1.3 for the tube size in these tests. In the air ambient, the flame made a gradual transition in intensity, becoming dimmer and larger as the equivalence ratio increased for the same outlet velocity. Most of the tests were run with the tube horizontal, and the flame appeared symmetric about the tube axis. As the flame became longer, the effect of buoyancy did begin to be obvious and the flame stretched vertically for phi = 4 for the velocities in this investigation. For certain conditions we discovered the flame would move inside the tube and begin slowly traveling upstream toward the fuel source. At first, the flame stabilizes on the tube end. It heats the end of the tube, and when that is sufficiently hot the flame is able to go inside. Then the flame continues preheating the tube and advancing upstream. As the flame moves away from the end of the tube, the tube downstream of the flame cools and is no longer visible. The flame releases enough heat to cause the tube wall to glow brightly, making the flame location easy to determine.

Task Progress:

The rate of flame propagation for flames in four different microtubes as a function of mixture equivalence ratio was measured. The upstream flow velocity is held constant at 2 m/s. The flame velocity was obtained by recording a standard video of the outside of the tube, which glowed very brightly in the flame location and then tracking the flame front using software developed at GRC. Once in the tube, all of the flames had a steady propagation rate, so the slope from a position vs. time graph yielded the velocity. Note that the tube diameters are all close to the conventional quenching diameter for phi = 0.8 (~ 500 micronmeters), but much smaller at phi = 1.2 (~700 micronmeters).

For tubes of the same material (alumina ceramic), the thicker walled tube yielded a higher flame propagation rate. There are two competing effects regarding conduction in the tube wall. Heat conducted radially is lost into the wall and eventually to the ambient, while that conducted axially assists in flame propagation. In this case, the thicker wall provides more cross-sectional area to conduct heat upstream, outweighing the effect of radial conduction. It is believed that this effect will only exist until a critical wall thickness is reached.

A high tube thermal conductivity also led to high flame propagation velocities. This is evident in the curve for the stainless steel tube. As the temperatures of the flame increases, the thermal conductivities of the alumina and stainless steel begin to depart. At high temperatures the stainless steel thermal conductivity increases while that of alumina decreases. At phi = 1.0, the fact that the wall thicknesses of the two tubes are similar is overwhelmed by the large difference in thermal conductivities. The flames in the ceramic tubes propagated with the highest velocities at an equivalence ratio just under 1.0. This may be due to kinetics, but may also be partly explained by the fact that the thermal conductivity of the ceramic Al2O3 achieves its lowest values at temperatures between 1300K and 1800 K. According to the model of the heat transfer in the tube, these are the temperatures the tube will be experiencing at an equivalence ratio of one. Lowering the temperature by decreasing the equivalence ratio below 1.0 causes the temperature of the tube to drop, but at the same time the thermal conductivity begins to increase. At these high temperatures a drop in temperature from 1300 K to 900 K could nearly double the thermal conductivity from approximately 4 to 8 W/mK. The drop in flame speed associated with a lean adjustment of the equivalence ratio may be overwhelmed by the increase in flame speed due to a large increase in thermal conductivity. It was also found that despite the fused silica having a lower thermal conductivity (1.4 W/mK), it supported higher propagation velocities than alumina. The reason for this is not obvious. More work is needed to untangle the effects of the wall thickness and material for cases where the thermophysical properties vary with temperature.

The silica tube has a transparent wall, allowing images to be obtained of a propagating flame. A numerical model developed by Lee and Tsai for larger tubes predicted either mushroom or tulip shaped flames, depending on the conditions. In these micro-tubes, the flame is neither, and takes on a fairly flat profile.

In contrast to the flame propagation above, at certain operating conditions, a flame stabilized just inside the outlet of a tube; the flame will travel neither upstream nor will it blow out the end. Such behavior was present with both ceramic and stainless steel tubes. With the fused silica tube rapid softening occurred and we could not test for a long enough time to stabilize the flame (some melting also occurred with the stainless steel tube, but not as rapidly). When the flame was inside the tube, no blue flame was visible outside, but the end of the tube glowed brightly. One mechanism of stabilization for the flames near the outlet of the tube is an asymmetry of heat losses. If a flame is in the middle of the tube, it can lose heat both upstream and downstream due to conduction in the tube wall. When the flame resides near the outlet of the tube, the major heat loss is upstream, since there is no tube downstream to act as a conduction path. This leads to an increase in the temperature of the tube near the outlet. The flame is able to heat the tube enough to propagate into it, but not enough to continue propagating upstream.

To better understand this effect, a numerical model of the heat transfer in the microtube was developed to predict tube wall temperatures, heat losses, etc. The steady state model accounts for axial conduction in the tube wall, radial convective and radiant heat losses to the environment, convective heat transfer to and from the flow, heat release in the gas via an energy source term based on complete combustion, and variable thermophysical and transport properties. The control volumes are defined using the SIMPLE algorithm and the coupled equations are solved iteratively. The effect of wall thickness in the model is apparent, with the thin walled tube getting hotter locally, but the temperature dropping off more quickly away from the region of heat release.

Further experiments with the microflames revealed it is also possible to manually force the flame to stabilize inside the tube. In certain conditions where the flame is close to entering the tube, but definitely stabilized outside the tube, one could manually preheat the tube with a match, which would cause the flame to move and stabilize inside the tube. Removing the match did not cause the flame to move back outside, indicating the presence of two stability points.

Fuel leakage underneath an exterior stabilized flame can lead to flame extinction and/or incomplete combustion. With the flame deep inside a microtube, however, much of the heat is lost due to conduction in the walls and subsequent radiation and convection to the environment. For a microcombustor application where outlet heat delivery is most important, it seems that having the flame stabilized just inside the end of the tube is optimal, as no fuel leakage can occur and the highest outlet temperature is at the end of tube.

At low velocities and slightly rich equivalence ratios, when the flame is stabilized inside the tube, significant acoustic emissions can be heard from the tubes. This sound tended to be of single frequency, and was loud enough to be heard through a closed door. Data acquired by the microphone was processed with a fast Fourier transform to obtain an amplitude vs. frequency plot. Typical frequencies obtained were about 1.8 kHz. Originally, we believed the emission was similar to that from an organ pipe, but experiments with different length tubes showed no frequency dependence on tube length. An investigation with the fused silica tube was performed to determine what the flame was doing inside the tube when sound was being emitted. The video showed that when the acoustic emission occurs, the flame appears to stretch out from a disk and fill a length of tube equal to many diameters, and the emitted frequency depended on the flame length. The flame appeared long in the video (captured at 30 fr/s), but this is probably due to the flame rapidly pulsing back and forth in the tube.

Regarding the numerical model, during this period we completed inclusion of the gas-phase reactions and added heat loss and axial conduction to the tube wall. The latter required a change in solution procedure from a marching scheme to one that accounts for downstream conditions (due to the axial conduction term). We have begun to run parametric cases with carbon monoxide and hydrogen to determine the effects of initial tube and gas temperature.

Bibliography Type: Description: (Last Updated: 06/15/2021)

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